

Unit #1 : Introduction to Bayesian Inference

Bayesian Statistical Methods, Chapter 1



Photo by Glenn Asakawa/University of Colorado



STOP

Photo by [Jana Knorr](#) on [Unsplash](#)

A **stopping rule** is a rule that tells a researcher when to conclude data collection.

Example: Thom is collecting data on coffee sales at a local coffee shop. The purpose of his analysis is to infer the how coffee sales vary by demographic (e.g., do younger or older customers spend more money at the coffee shop?). He decides to observe coffee sales *until he's finished with his cup of coffee*.



STOP

Imagine health researchers are investigating the prevalence of a rare genetic condition, denoted as G . If the percentage of individuals carrying the G gene in the population, p , is greater than 0.5, certain recommendations might be given to insurance companies to cover further screenings for G . The researchers set the significance level at $\alpha = 0.1$.

They collected the following binary data, where 1 = "does not carry G " and 0 = "carries G ":

$$\mathbf{x} = (0,0,0,1,0,1,0,0,0,1)$$

What was the stopping rule?



STOP

Activity: Should “stopping rules” matter? Do they matter in frequentist inference?

An abstract light painting on a dark background, featuring a bright orange-yellow spiral that winds upwards and outwards, creating a sense of motion and depth. The spiral is composed of many small, overlapping light trails, giving it a textured, almost 3D appearance. The background is a deep, dark brown or black, which makes the bright spiral stand out prominently.

Probability and statistics review:

- Basic theorems of probability (including Bayes' theorem for events!).
- Univariate distributions/random variables (both discrete and continuous).
- **Multivariate distributions.**
- Random samples, distributions of estimators/test statistics, the central limit theorem.
- Estimation theory (maximum likelihood estimation, properties of estimators)
- Confidence intervals.
- Hypothesis testing.



An n -dimensional vector $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ as a **multivariate normal distribution** with mean vector $E(\mathbf{Y}) = \boldsymbol{\mu}$ and variance covariance matrix $Var(\mathbf{Y}) = \Sigma$ if, for any constant (i.e., *non-random*) vector $\mathbf{a} = (a_1, a_2, \dots, a_n)^T$, every linear combination $X = \mathbf{a}^T \mathbf{Y} = a_1 Y_1 + a_2 Y_2 + \dots + a_n Y_n$ has a univariate normal distribution with mean $\mu_X = \mathbf{a}^T \boldsymbol{\mu}$ and variance $\sigma_X^2 = \mathbf{a}^T \Sigma \mathbf{a}$.



An n -dimensional multivariate normal vector $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ can be written in the following notation: $\mathbf{Y} \sim N(\boldsymbol{\mu}, \Sigma)$, where $\boldsymbol{\mu} = (\underbrace{\mu_1}_{E(Y_1)}, \dots, \underbrace{\mu_n}_{E(Y_n)})^T$ is a vector of means and variance-covariance matrix given by

$$\Sigma = \begin{pmatrix} \sigma_{1,1} & \sigma_{1,2} & \cdots & \sigma_{1,n} \\ \sigma_{2,1} & \sigma_{2,2} & \cdots & \sigma_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{n,1} & \sigma_{n,2} & \cdots & \sigma_{n,n} \end{pmatrix}.$$

The entries of Σ are:

$$\sigma_{i,j} = \text{Cov}(Y_i, Y_j) \text{ for } i, j = 1, \dots, n.$$

An abstract photograph featuring several bright, golden-yellow light trails against a dark, textured background. The trails are composed of many small, closely spaced points of light, creating a sense of motion and depth. Some trails are straight and vertical, while others are curved and horizontal, intersecting to form a complex, layered pattern. The overall effect is reminiscent of long-exposure photography of light sources like stars or distant galaxies.

Example: Let $\mathbf{Y} = (Y_1, \dots, Y_n)^T$ be an n -dimensional multivariate normal vector:
 $\mathbf{Y} \sim N(\boldsymbol{\mu}, \Sigma)$, where $\boldsymbol{\mu} = (\mu_1, \dots, \mu_n)^T$ is a vector of means and variance-covariance matrix given by Σ .

The background of the slide features an abstract, artistic composition. It consists of several thin, glowing orange lines that appear to be light trails or fiber optics. These lines are set against a dark, almost black, background. The lines are mostly vertical, but some curve and loop, creating a sense of movement and depth. The overall effect is reminiscent of a long-exposure photograph of light or a digital art piece.

Conditional distributions:

The conditional distribution of X_1 given that the random variable X_2 is fixed at x_2 is given by:

$$f_{X_1 | X_2} \left(x_1 \mid X_2 = x_2 \right) = \frac{f(x_1, x_2)}{f(x_2)}, \quad f(x_2) > 0.$$



Conditional from multivariate normal:

Let $(Y_1, Y_2) \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$.

Let $\rho = \frac{\sigma_{1,2}}{\sqrt{\sigma_{1,1}\sigma_{2,2}}} = \frac{\text{Cov}(Y_1, Y_2)}{\sqrt{\text{Var}(Y_1)\text{Var}(Y_2)}}$. Then

the distribution $Y_2 \mid Y_1 = y_1$ is given by

$$Y_2 \mid Y_1 \sim N\left(\mu_2 + \rho\sqrt{\frac{\sigma_{22}}{\sigma_{11}}}(y_1 - \mu_1), (1 - \rho^2)\sigma_{22}\right)$$

The left side of the slide features a vertical strip of abstract light trails. These trails are composed of numerous thin, overlapping lines of warm, golden-yellow light. Some lines are straight and vertical, while others are curved, looping, or spiral-like, creating a sense of movement and depth against a dark, textured background.

Let (Y_1, Y_2) be bivariate normal with
 $E(Y_1) = E(Y_2) = 0$, $Var(Y_1) = Var(Y_2) = 1$,
and $Cov(Y_1, Y_2) = \rho$.

1. What is the distribution of Y_1 ?

2. Derive the conditional distribution of $Y_1 | Y_2$.



How is probability theory used in statistical inference?

An abstract photograph featuring several bright, golden-yellow light trails against a dark, textured background. The trails are composed of many small, closely spaced points of light, creating a sense of motion and depth. Some trails are straight and vertical, while others are curved, looping, and horizontal, intersecting in a complex, organic pattern. The overall effect is reminiscent of long-exposure photography of light sources or perhaps a microscopic view of certain biological structures.

“Should probability enter [statistics] to capture degrees of belief about claims? To measure variability? Or to ensure we won’t reach mistaken interpretations of data too often in the long run of experience? Modern statistical methods grew out of attempts to systematize doing all of these.”

—Deborah Mayo, *Statistical Inference as Severe Testing*

An abstract photograph featuring several bright, golden-yellow light trails against a dark, textured background. The trails are composed of many small, closely spaced points of light, creating a sense of motion and depth. Some trails are straight and vertical, while others curve and loop, resembling a spiral or a complex path. The overall effect is one of dynamic energy and intricate patterns.

“The field [of statistics] has been marked by disagreements between competing ... [groups] of frequentists and Bayesians that have been so contentious – likened in some quarters to religious and political debates – that everyone wants to believe we are long past them...The truth is, long-standing battles still simmer below the surface in questions about scientific trustworthiness and the relationships between Big Data-driven models and theory.”

—Deborah Mayo, *Statistical Inference as Severe Testing*



What is frequentist statistical inference?

The image features a dark, textured background with several bright, orange-yellow light trails. These trails are composed of many small, closely spaced dots or segments, creating a sense of motion and depth. Some trails are straight and vertical, while others are curved, looping, or spiral-like, suggesting a complex, possibly mathematical or physical, process. The overall effect is one of dynamic energy and abstract beauty.

“By and large, classical statistical procedures share the feature that they only rely on probability assignments over sample spaces...an important motivation for this is that those probabilities can be interpreted as frequencies.” —Jan-Willem Romeijn, <https://stanford.io/3JJD6rS>

A frequentist statistical model

$$(\mathbf{X}, f(\mathbf{x}; \theta))$$

Random variables that generate the data.

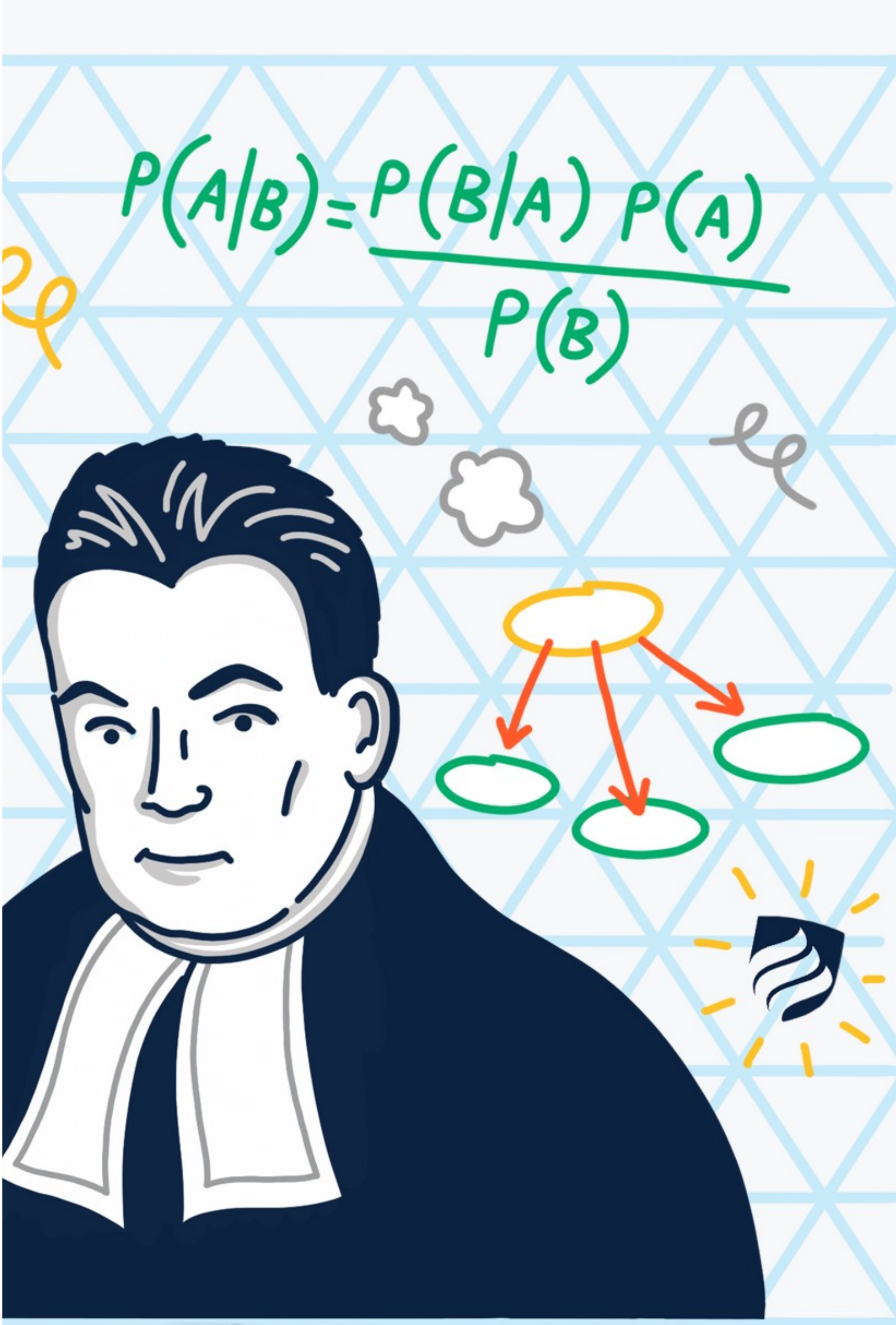
The joint probability distribution over the random variables, for a fixed set of parameters, θ .

“The *statistical model* is chosen to represent a description of a chance mechanism that accounts for the systematic information (the chance regularities) in the data.” —Aris Spanos

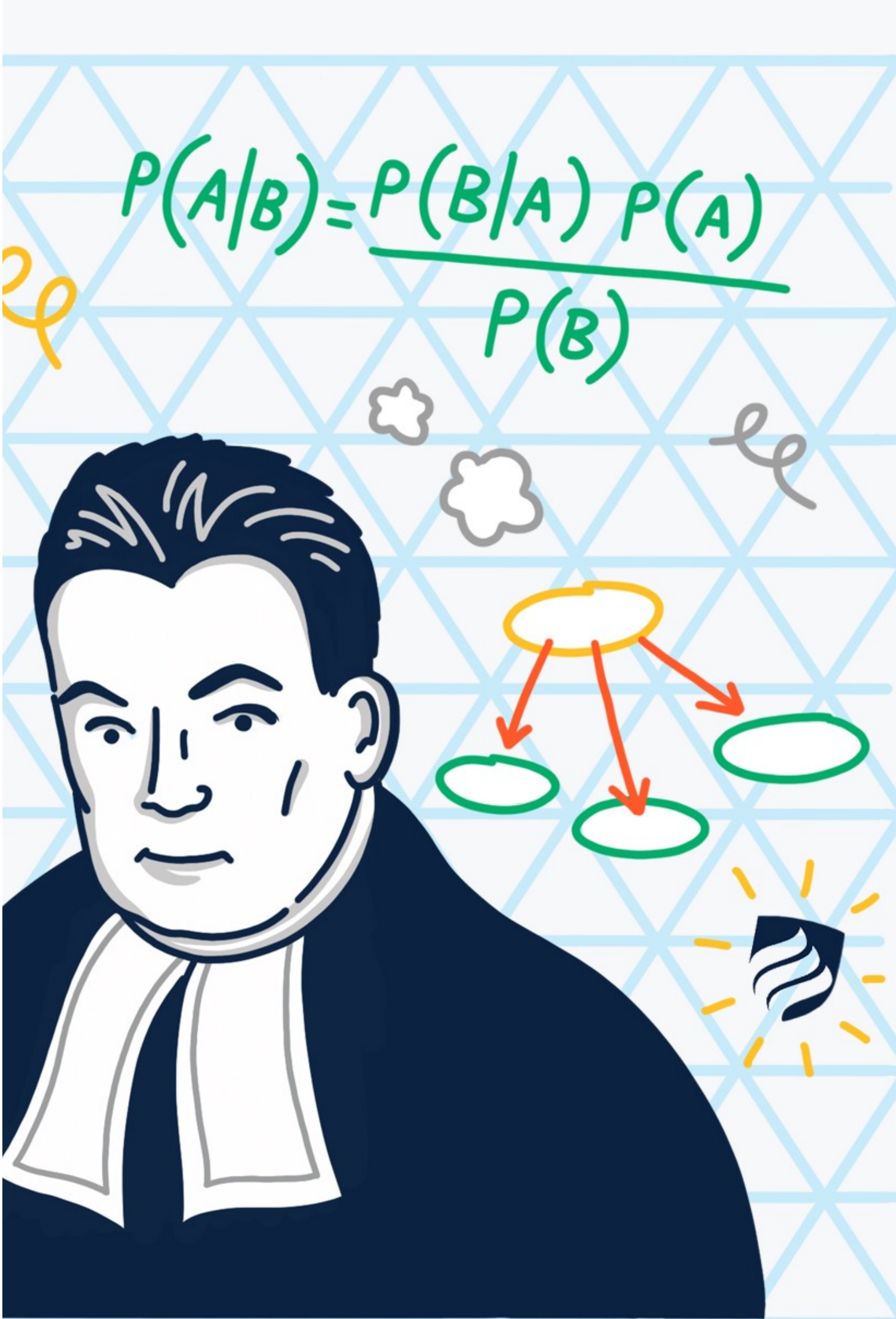
An abstract photograph featuring several bright, golden-yellow light trails against a dark, textured background. The trails are composed of many small, closely spaced points of light, creating a sense of motion and depth. Some trails are straight and vertical, while others are curved and horizontal, intersecting to form a complex, geometric pattern. The overall effect is reminiscent of long-exposure photography of light sources or perhaps a stylized representation of data paths.

Frequentist methods in statistics:

1. *Hypothesis testing*. z-test, t-test, chi-squared test, F-test.
2. *Point estimation*. Maximum likelihood estimation (in one context, least squares), method of moments.
3. *Interval estimation*. Confidence intervals.



What is a Bayesian statistics?



“By and large, classical statistical procedures share the feature that they only rely on probability assignments over sample spaces...an important motivation for this is that those probabilities can be interpreted as frequencies.” — Jan-Willem Romeijn, <https://stanford.io/3JJD6rS>

“The defining characteristic of Bayesian statistics is that it considers probability distributions over statistical hypotheses as well as over data.”—Jan-Willem Romeijn, <https://stanford.io/3qlarKQ>



A Bayesian statistical model

$$(\mathbf{X}, f(\mathbf{x}, \theta))$$

Random variables that
generate the data

Joint probability
distribution over the data
and parameters



$$P(A/B) = \frac{P(B/A) P(A)}{P(B)}$$

Bayesian methods in statistics:

1. *Posterior distributions for a parameter (or set of parameters).* Computed using Bayes' theorem.
2. *Point estimation.* Uses the posterior distribution to find a single value to estimate an unknown parameter.
3. *Interval estimation.* Uses the posterior distribution to construct a range of values that, with some probability, contains the unknown parameter.
4. *Bayes' factors.* Provide a method for testing hypotheses.

Comparisons of frequentist and Bayesian statistics

Frequentist:

- Parameters are unknown and fixed.
- Probability theory describes the processes that generate data, and is *not* applied to parameters/hypotheses.
- The chance processes that generate data are used to estimate parameters or make inferences about hypotheses related to those parameters.
- We assess the *frequentist properties* of estimators and test statistics. For example, is the estimator unbiased? Consistent? Efficient?
- These properties have to do w/*frequency*, i.e., repeated sampling.
 - Example: if I chose a sample of size n over and over from this population, would my estimator be correct, on average?

Bayesian:

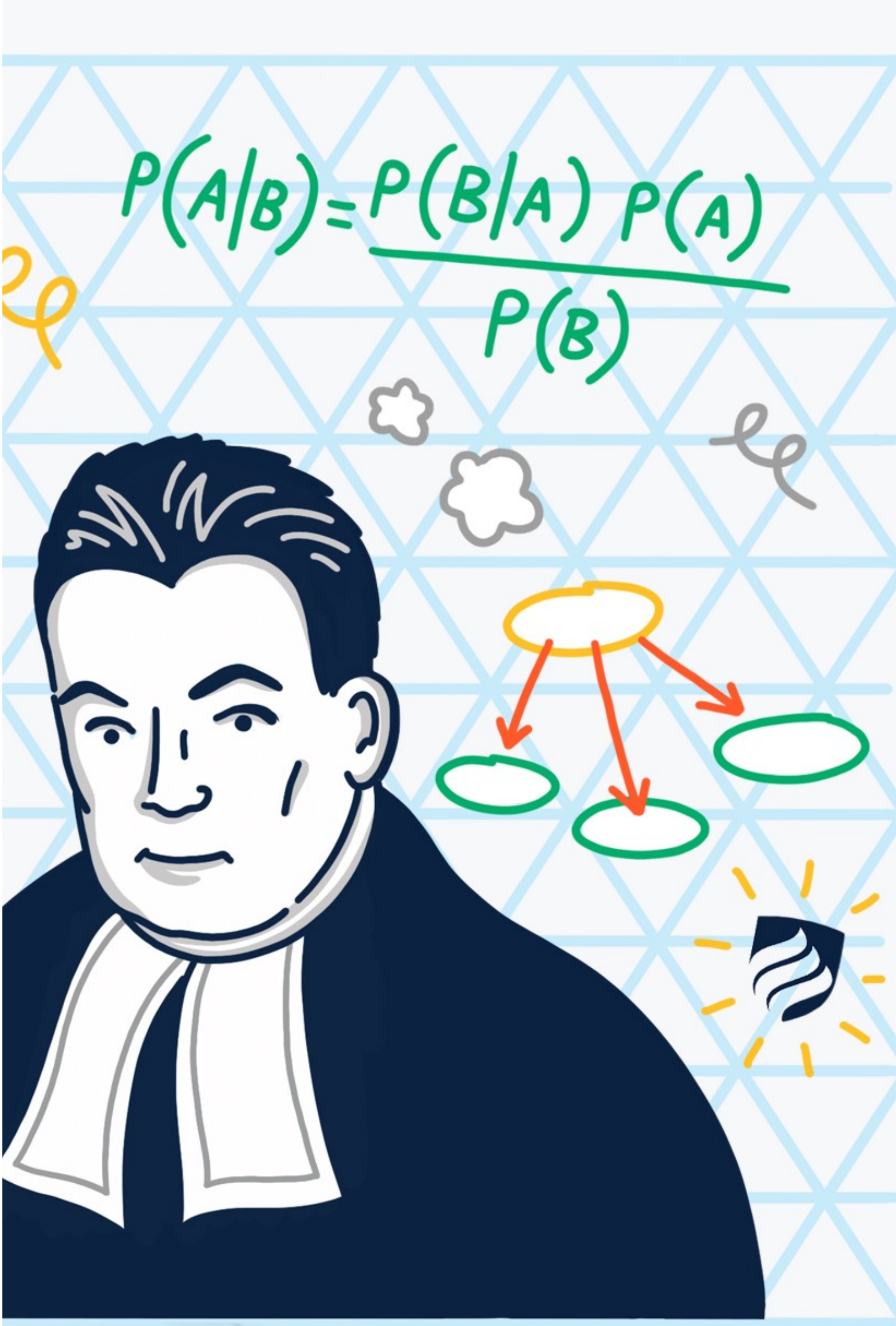
- Parameters are unknown and fixed, but we *model* them as random variables.
 - This model is usually thought of as representing our *degree of belief* in the value of the parameter.
 - We might think (before sampling): I don't know the (fixed) mean height of all CU students, but I believe that it's definitely between 5 and 6 ft. (Prior Belief).
- We use Bayes' theorem to combine our prior belief about a parameter with evidence from a sample.
- The result is a new probability distribution over the parameter. It represents our belief in the value of the parameter after accounting for evidence (the sample). The mean or mode might serve as an estimator..
- We can assess the frequentist performance (e.g., unbiasedness) of a Bayesian estimator.



$$P(A/B) = \frac{P(B/A) P(A)}{P(B)}$$

In the frequentist approach to estimation, hypotheses, represented statistically by model parameters, θ , were thought to be unknown and fixed (i.e., not random).

In the Bayesian approach, parameters are considered to be a quantities whose variation can be described by a (subjective) probability distribution (called a prior distribution). A sample is then taken from the population and the prior is updated—using Bayes' theorem—to obtain a posterior distribution.



From: <https://bit.ly/3gTjIBg>

$$(X, f(\mathbf{x}, \theta))$$

Random variables that
generate the data

Joint probability
distribution over the data
and parameters



$$(\mathbf{X}, f(\mathbf{x}, \theta))$$

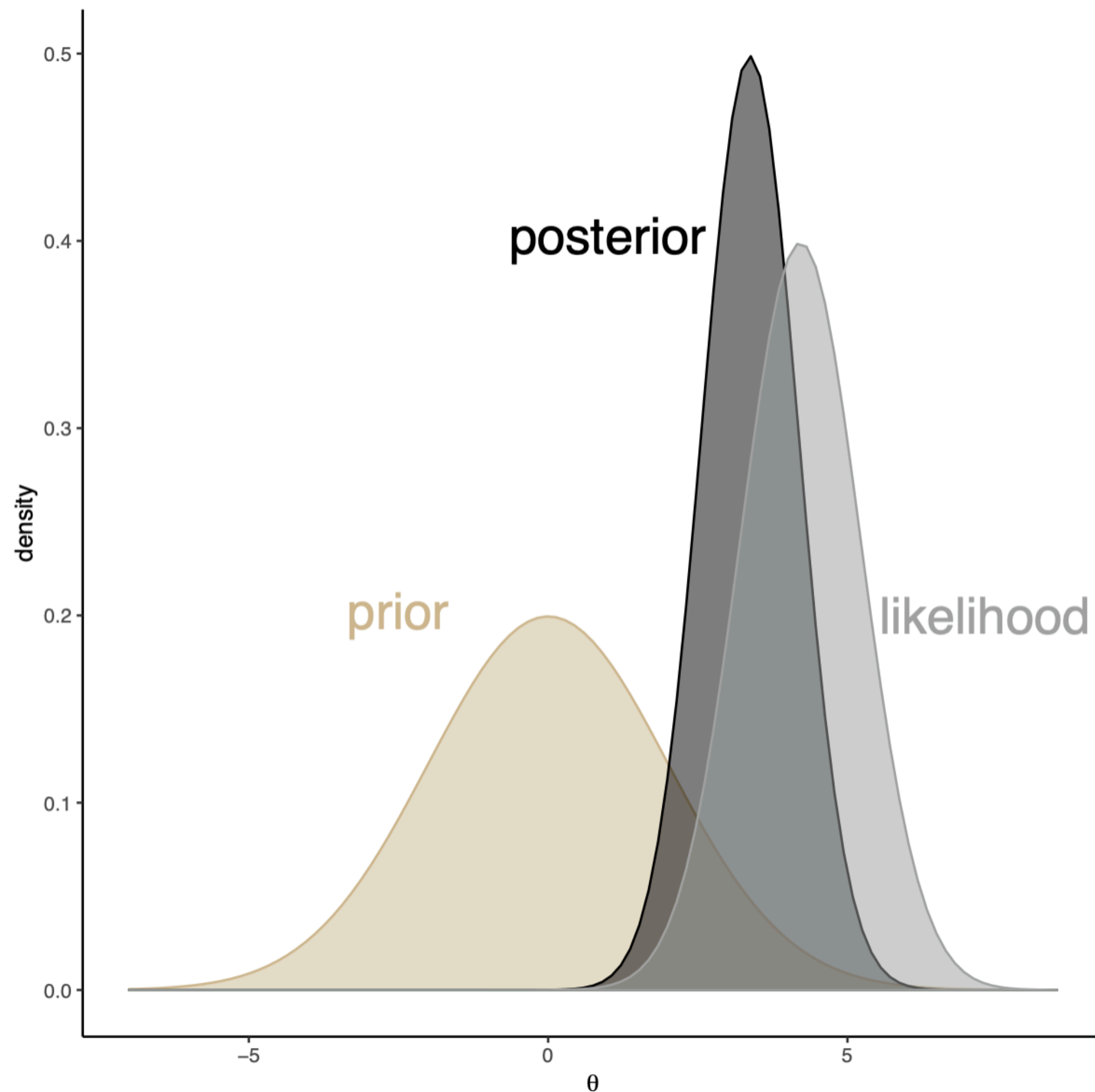
Random variables that
generate the data

Joint probability
distribution over the data
and parameters

$f(\mathbf{x}, \theta)$ is the joint probability of the data, $\mathbf{X}$, and the parameters, θ .

Denote the marginal distribution of θ as $\pi(\theta)$, the marginal distribution of $\mathbf{X}$ as $f(\mathbf{x})$, and the conditional distributions as $\pi(\theta | \mathbf{x})$ and $f(\mathbf{x} | \theta)$.

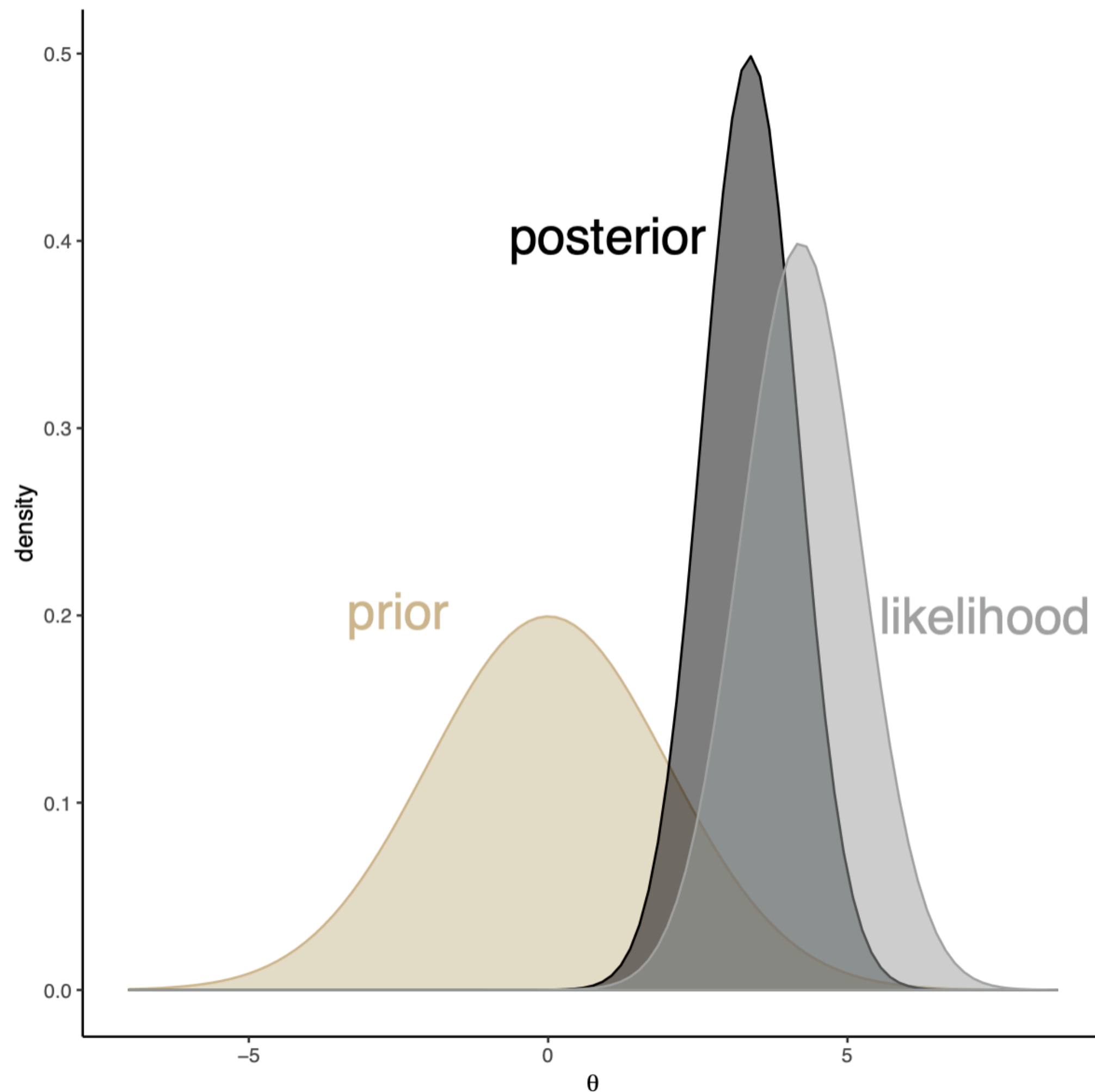
Then:



The marginal distribution of θ , $\pi(\theta)$, is called the **prior distribution**.

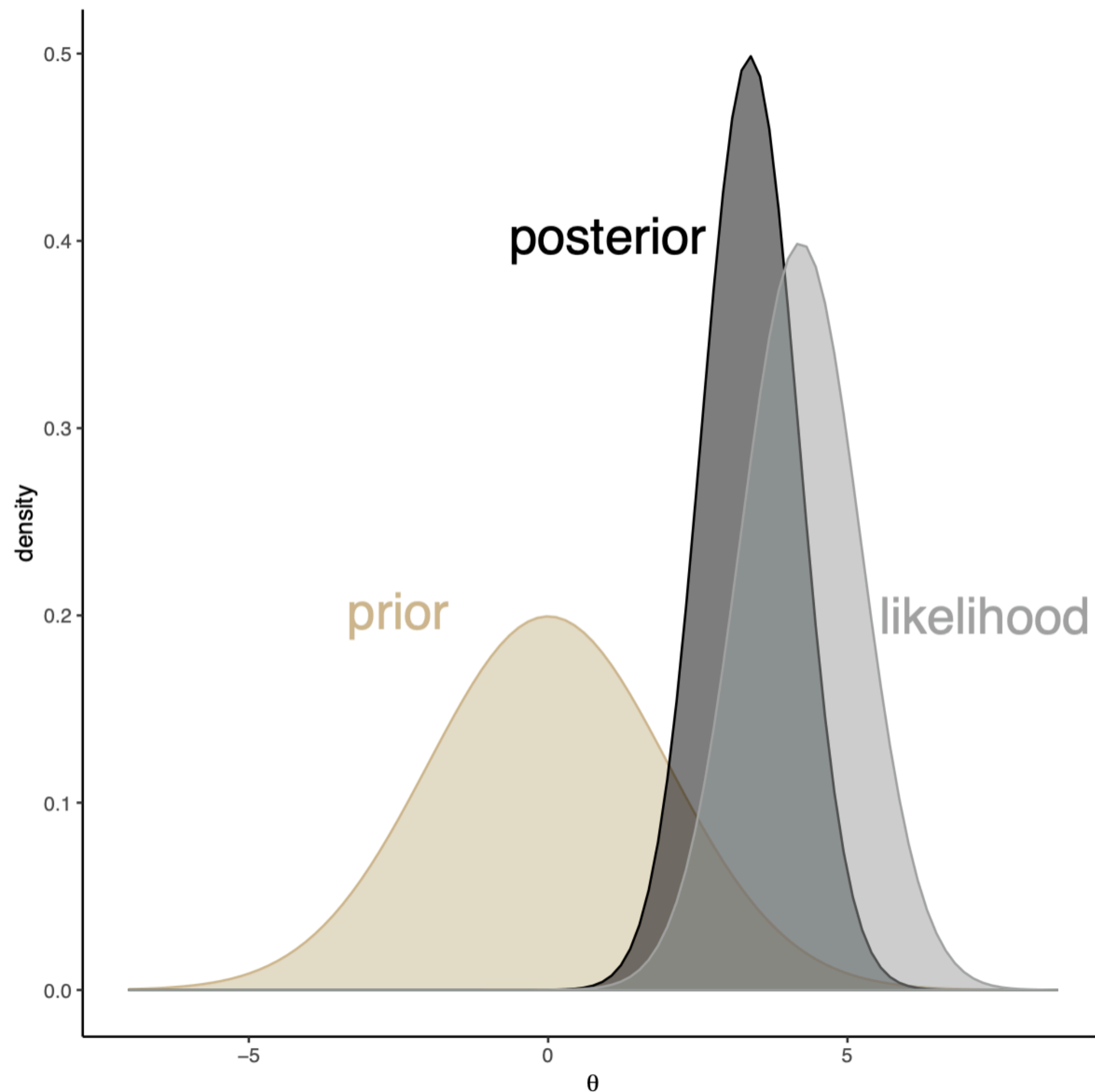
The conditional distribution of θ given the sample $\mathbf{x}$, $\pi(\theta | \mathbf{x})$, is called the **posterior distribution**.

The conditional distribution of $\mathbf{x}$ given the parameters θ , $f(\mathbf{x} | \theta)$, is called the **likelihood function**.



Notes on the prior distribution:

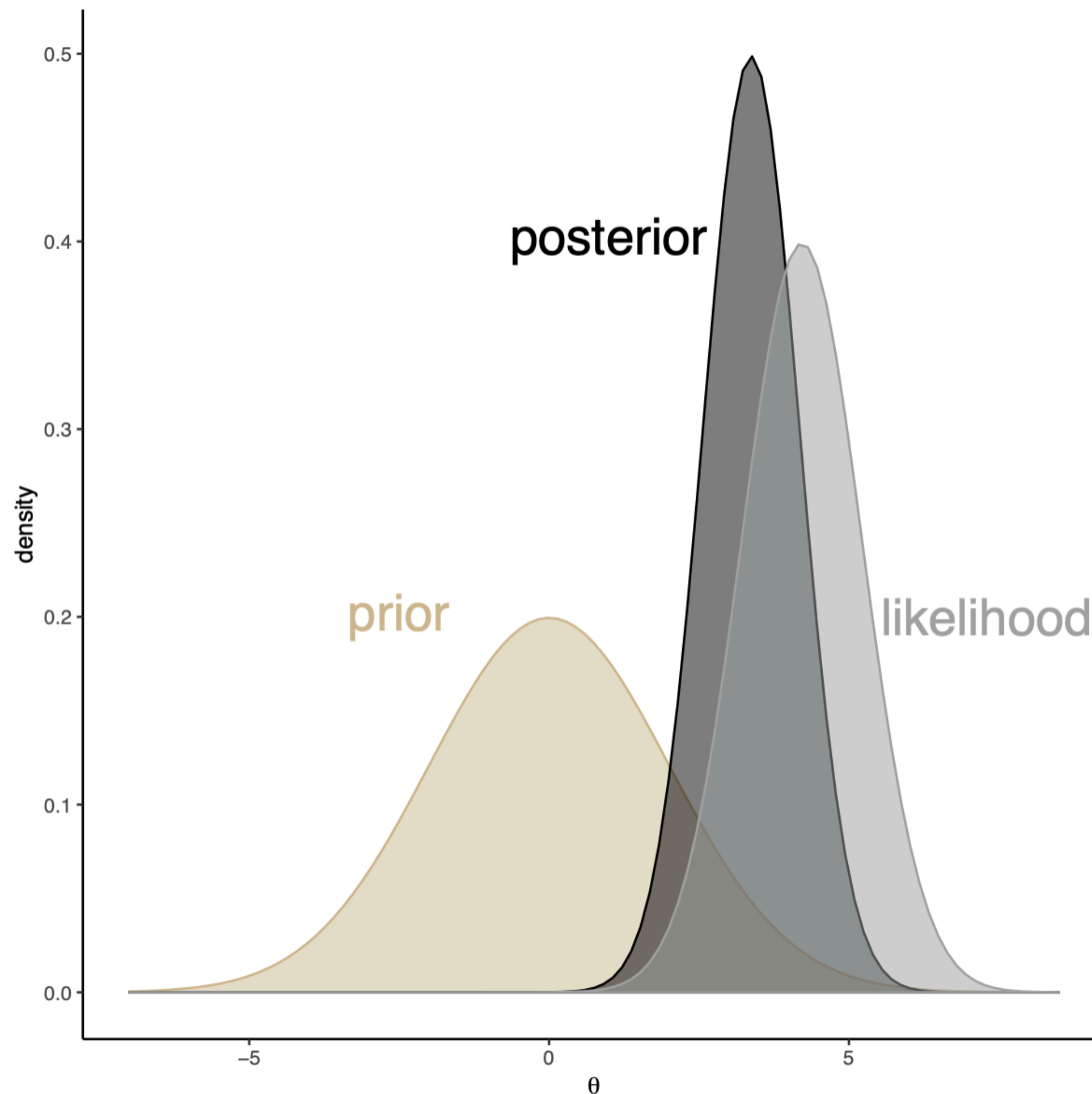
1. The prior distribution is meant to reflect our knowledge/degree of belief in the value of θ before any new data are observed.
2. It is often the case that we have prior knowledge, an expert opinion, etc. $\pi(\theta)$ is the formal way to take that knowledge into account.
3. There is a *ton* to say about what to do in the absence of prior information!
 1. "Uninformative" priors?
 2. "Objective" priors?



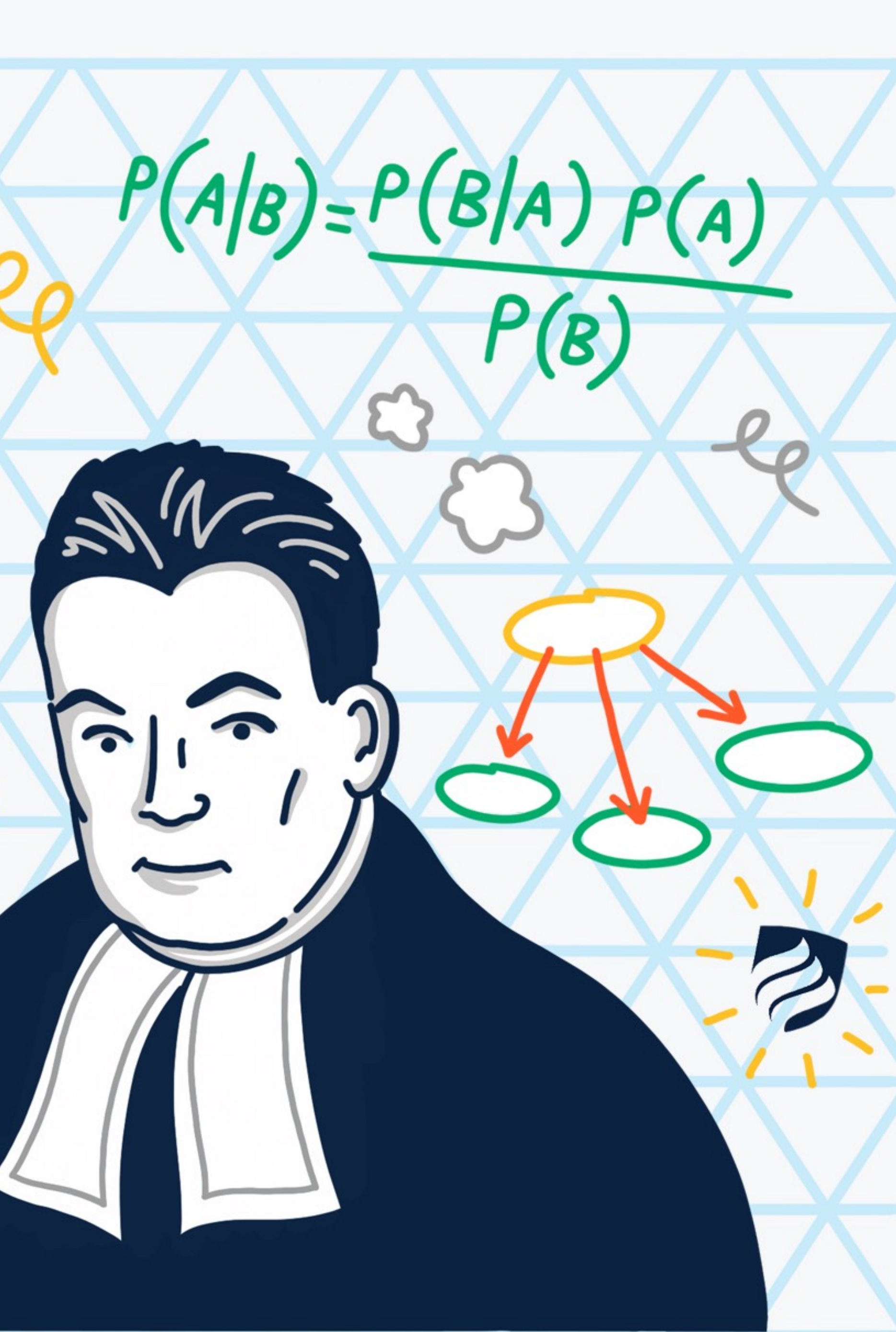
Notes on the likelihood function:

1. The Bayesian likelihood function is (almost) conceptually the same as the frequentist likelihood function (e.g., in MLE).
2. The likelihood gives the distribution of the sample given the parameter(s), but interpreted as a function of the parameters.

Notes on the posterior distribution:

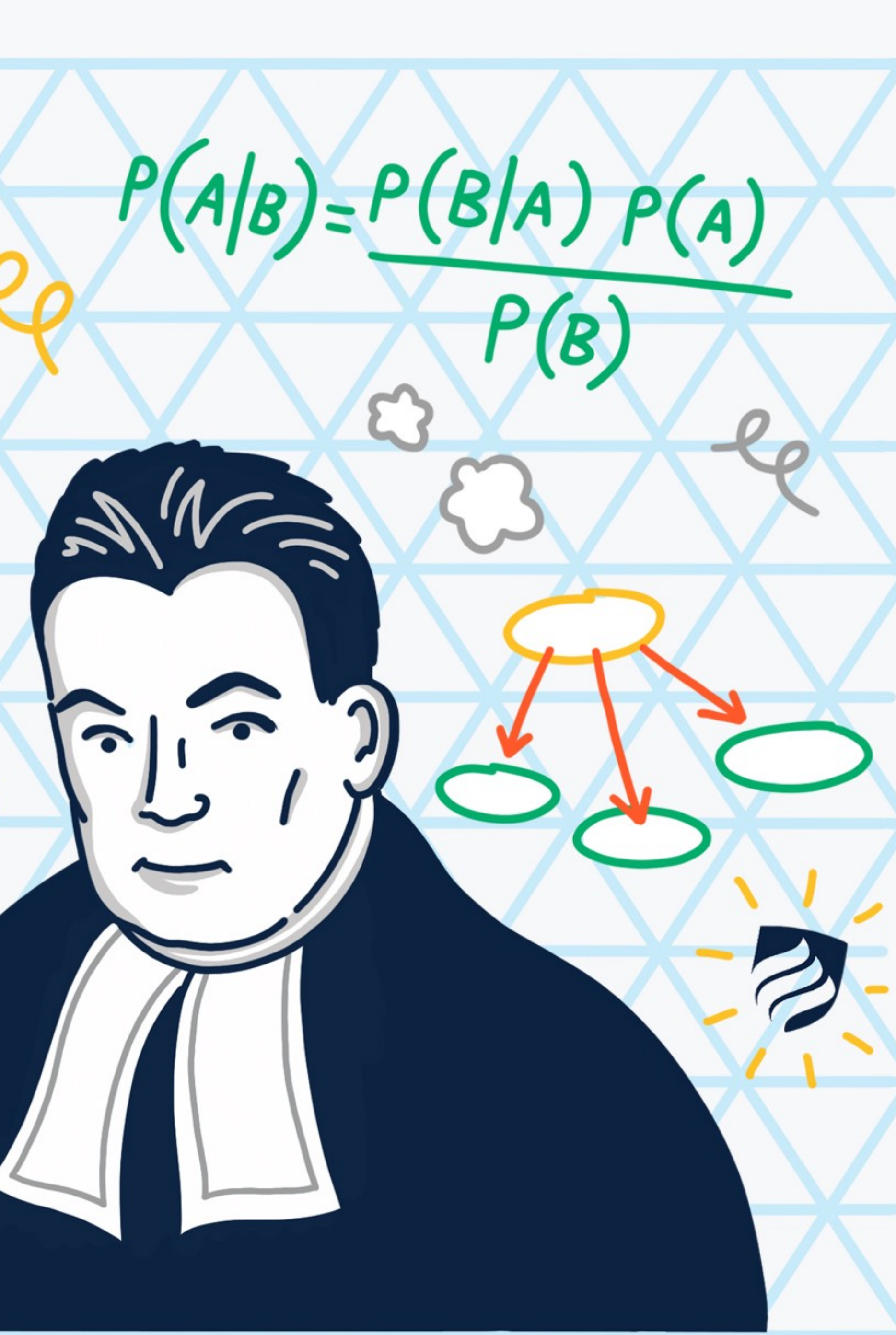


1. The posterior distribution describes our uncertainty about the parameter(s) given the data (and our prior knowledge).
2. It is used to make inferential statements about θ , which is still considered a random quantity. In Bayesian inference, an inference about θ is only valid if it is made based on the posterior distribution.
3. We can construct estimators/estimates of θ using the posterior, e.g., the mean or mode of the posterior can be used as a point estimate of θ .
4. We can use the posterior to construct interval estimates of θ .



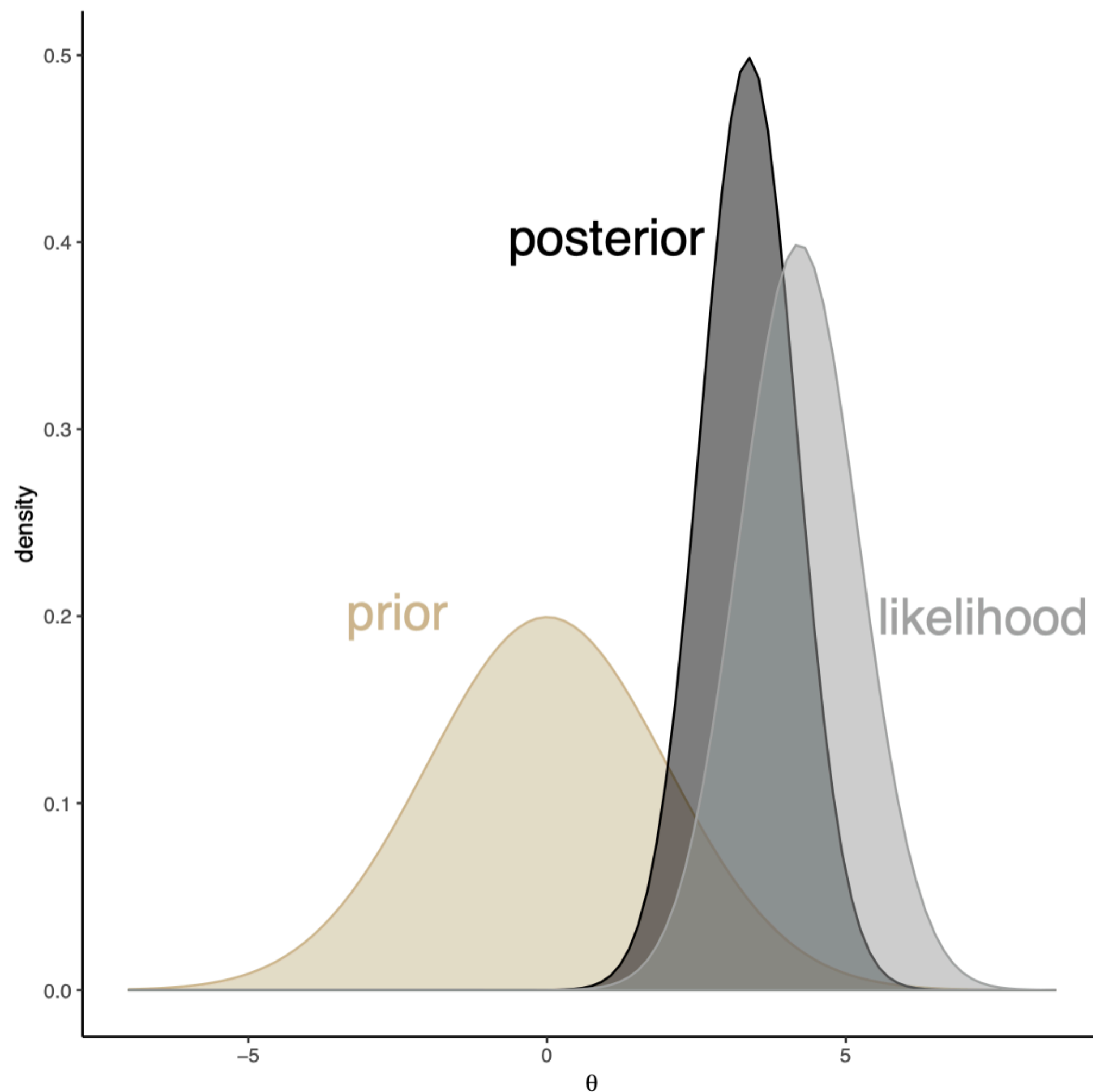
A note on the denominator of Bayes' theorem:

1. It is often very difficult to compute $f(\mathbf{x})$ directly, which is why we attempt to compute $f(\mathbf{x}) = \int f(\mathbf{x} | \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta}$ (or $f(\mathbf{x}) = \sum_{\boldsymbol{\theta}} f(\mathbf{x} | \boldsymbol{\theta}) \pi(\boldsymbol{\theta})$ when $\boldsymbol{\theta}$ is discrete).
2. $f(\mathbf{x}) = \int f(\mathbf{x} | \boldsymbol{\theta}) \pi(\boldsymbol{\theta}) d\boldsymbol{\theta}$ itself can be difficult to compute, especially when $\boldsymbol{\theta}$ is high-dimensional (e.g., imagine a complicated statistical learning model with hundreds or thousands of parameters...that's a tough integral!).
3. We will study techniques for avoiding the computation of this integral.



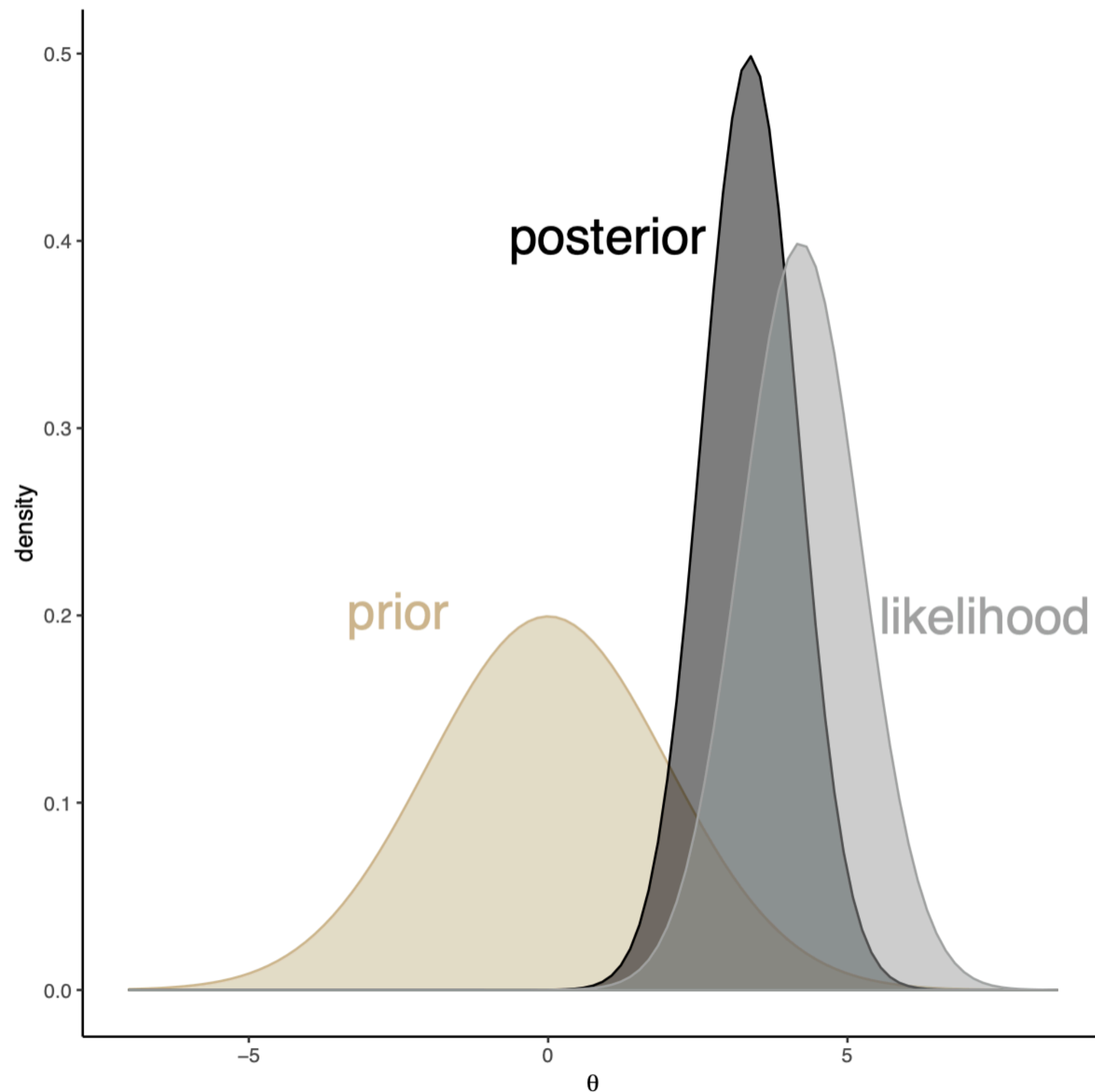
Activity: Open “Unit #1: Introduction to Bayesian Inference” from Canvas and work in groups of 2-3 on:

1. Problem #1(f) (continuing our discussion of stopping rules).
2. Problem #2 (a discrete example of Bayesian inference, and a connection to maximum likelihood).



Suppose that we have a posterior distribution, $\pi(\boldsymbol{\theta} \mid \mathbf{x})$.

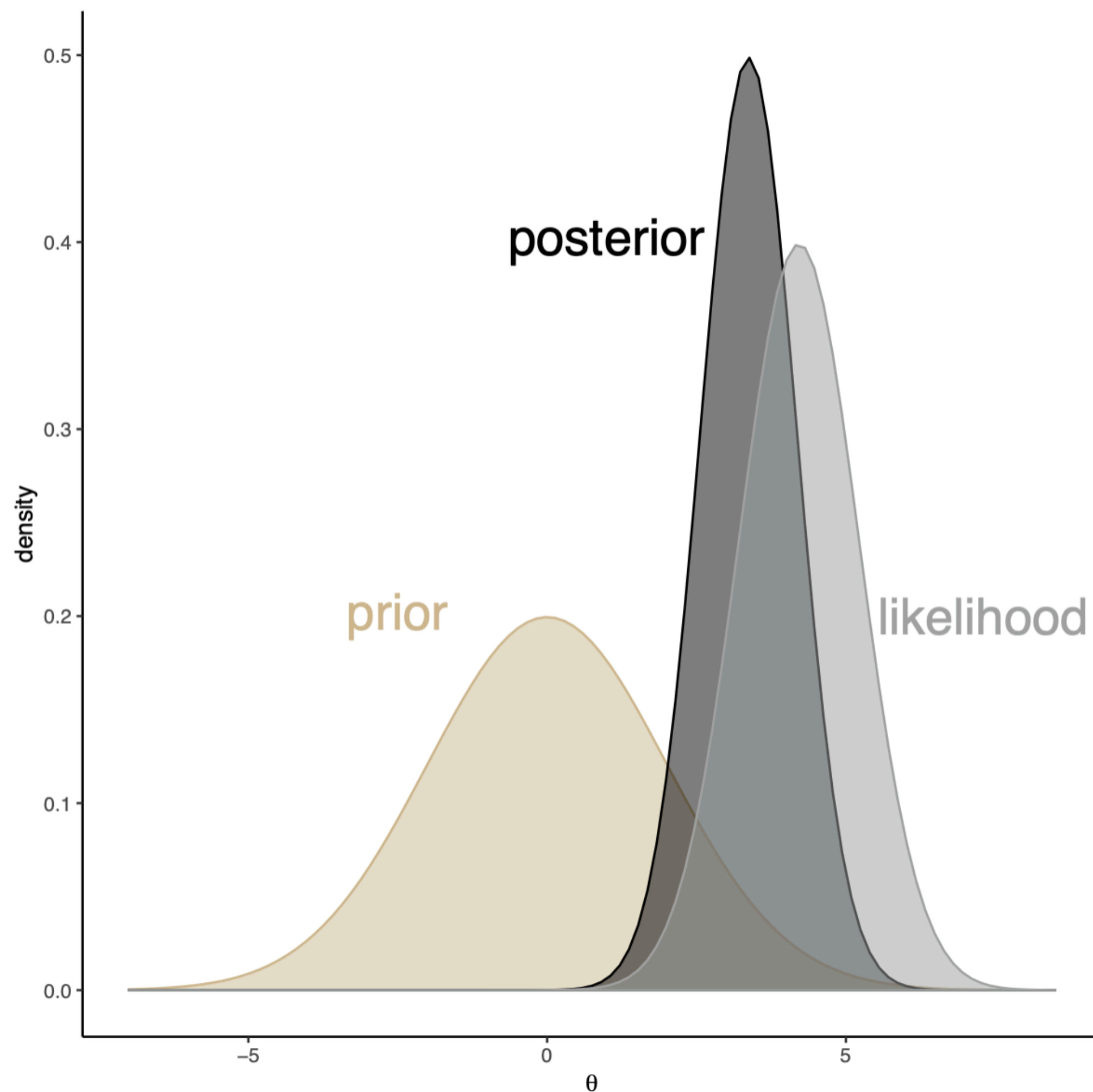
How can we summarize it? Which summaries will constitute reasonable estimators of $\boldsymbol{\theta}$?



Suppose that we have a posterior distribution, $\pi(\boldsymbol{\theta} \mid \mathbf{x})$.

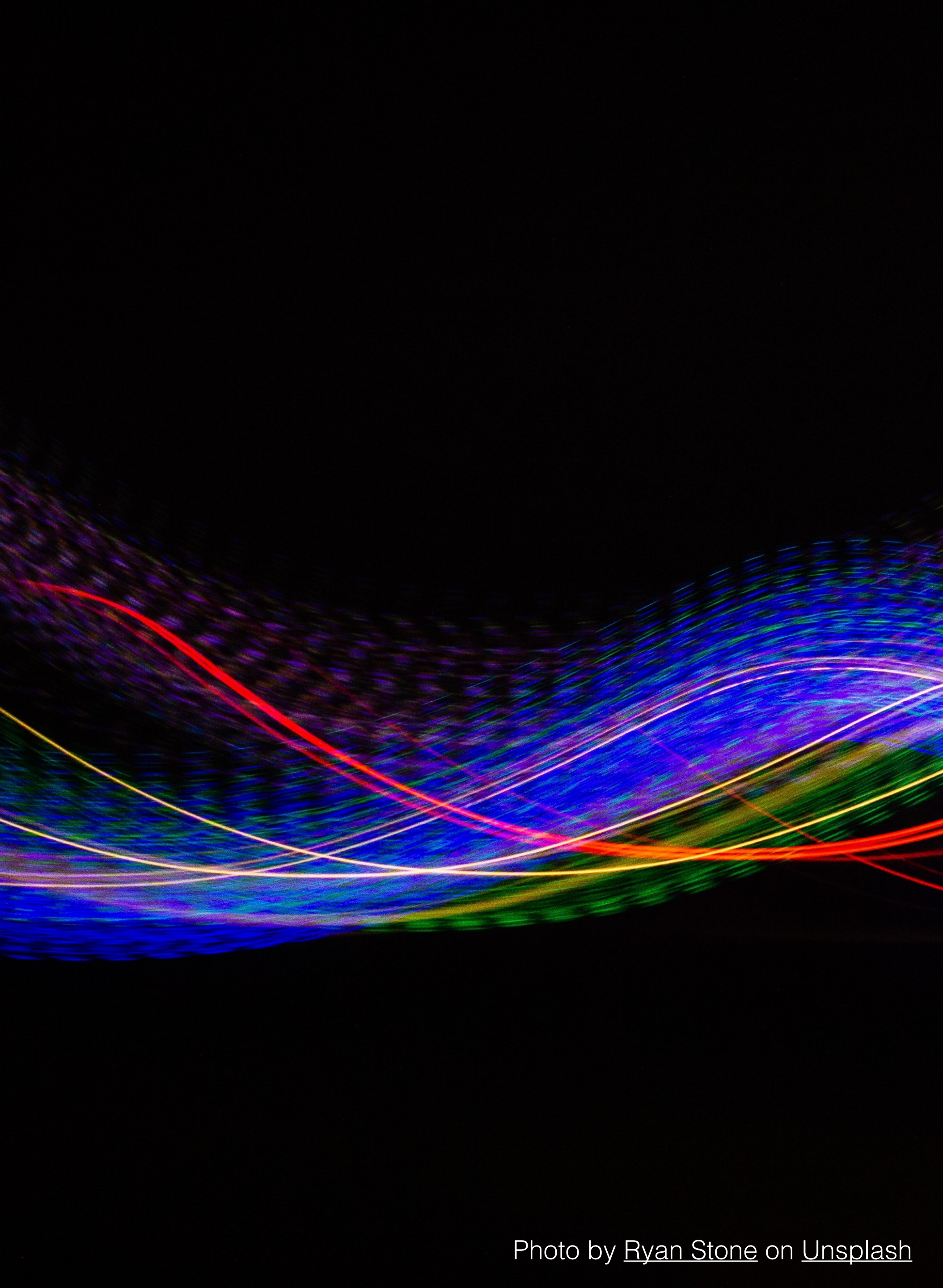
1. The **maximum a posteriori (MAP) estimator**:

2. The **mean/expectation of the posterior distribution**:

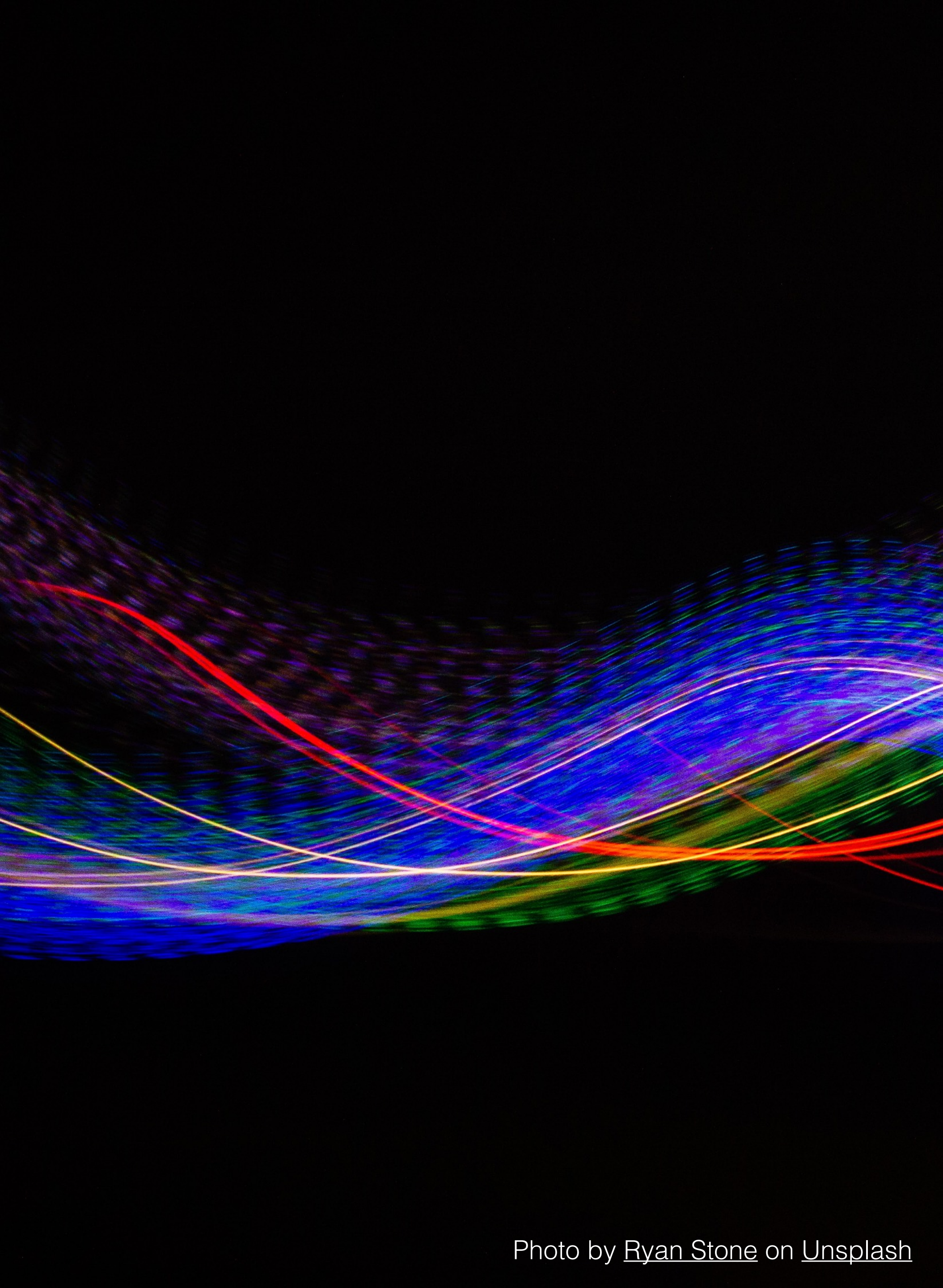


Suppose that we have a posterior distribution, $\pi(\theta \mid \mathbf{x})$, and we want an interval that we are **95 %** certain contains the true parameter, θ , given the data.

Let $Q(\theta)$ be the cdf of $\theta \mid \mathbf{x}$ and $Q^{-1}(x)$ be the inverse cdf of $\theta \mid \mathbf{x}$. Then we can write a **95 % credible interval estimator** as:

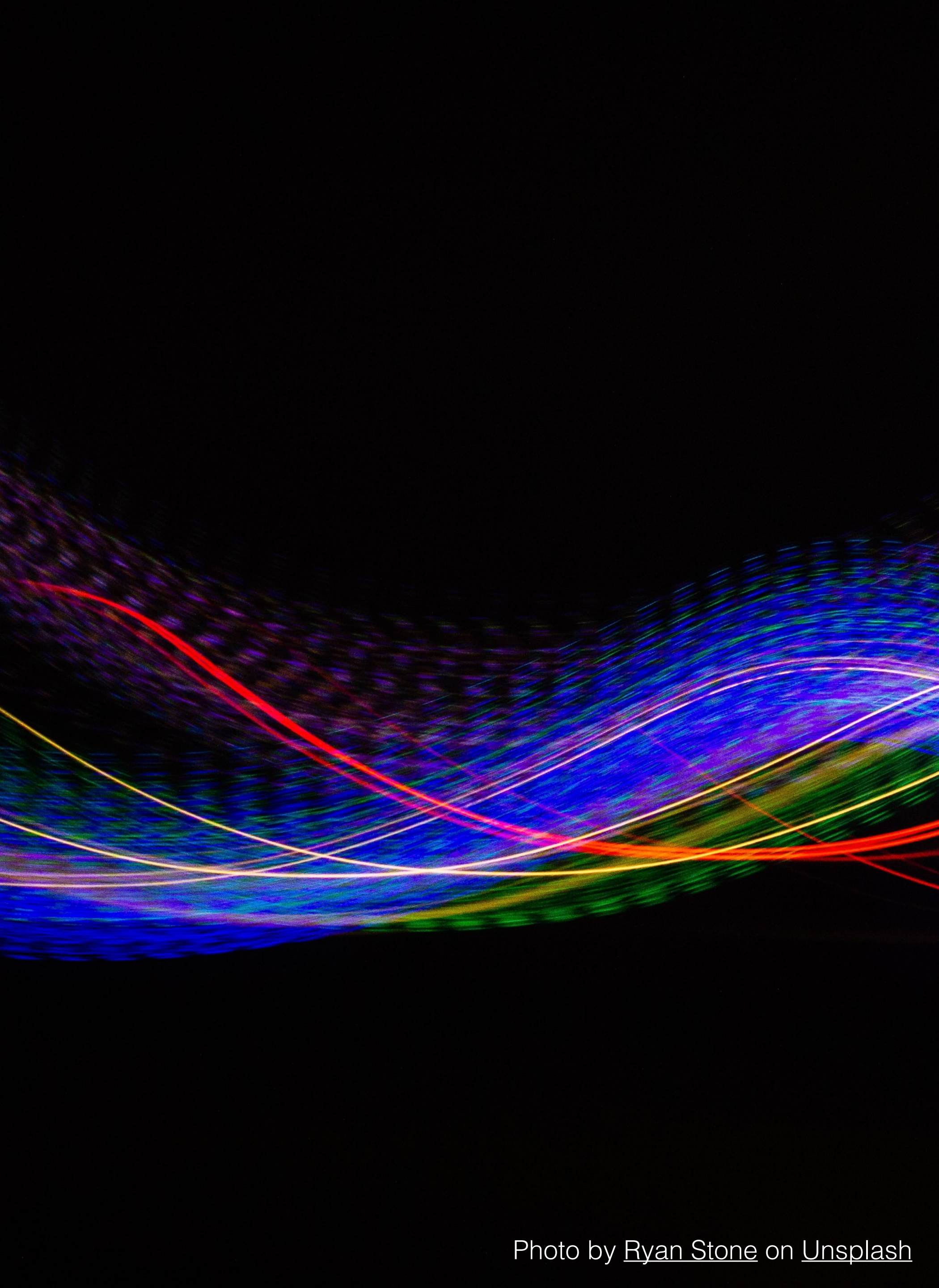
The left side of the image features a vertical black bar containing abstract, colorful light trails. These trails are composed of numerous thin, overlapping lines in shades of red, orange, yellow, green, blue, and purple, creating a sense of motion and depth. The trails are more concentrated in the lower half of the bar, with some lines extending upwards.

The goal of **explanatory modeling** is to make use of statistical models for testing and providing causal explanations.



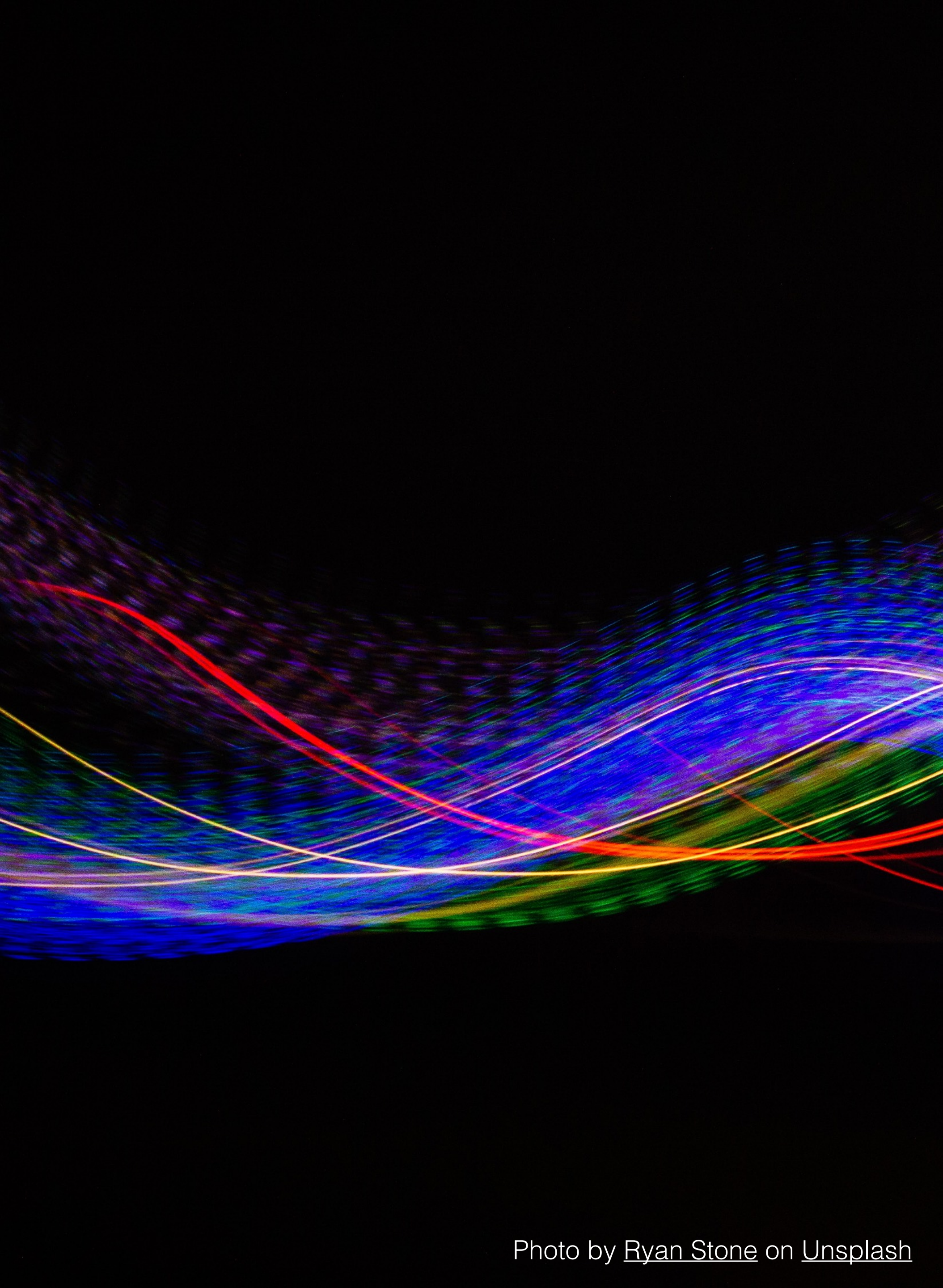
Explanatory models often:

1. Have parameters with a **physical meaning** (e.g., Hooke's Law).
2. Arise in the context of an **experiment design**.
3. Require additional assumptions if the data come from an **observational study**.



A **prediction** is an empirical consequence of a theory that has not yet been verified at the time the theory was constructed.

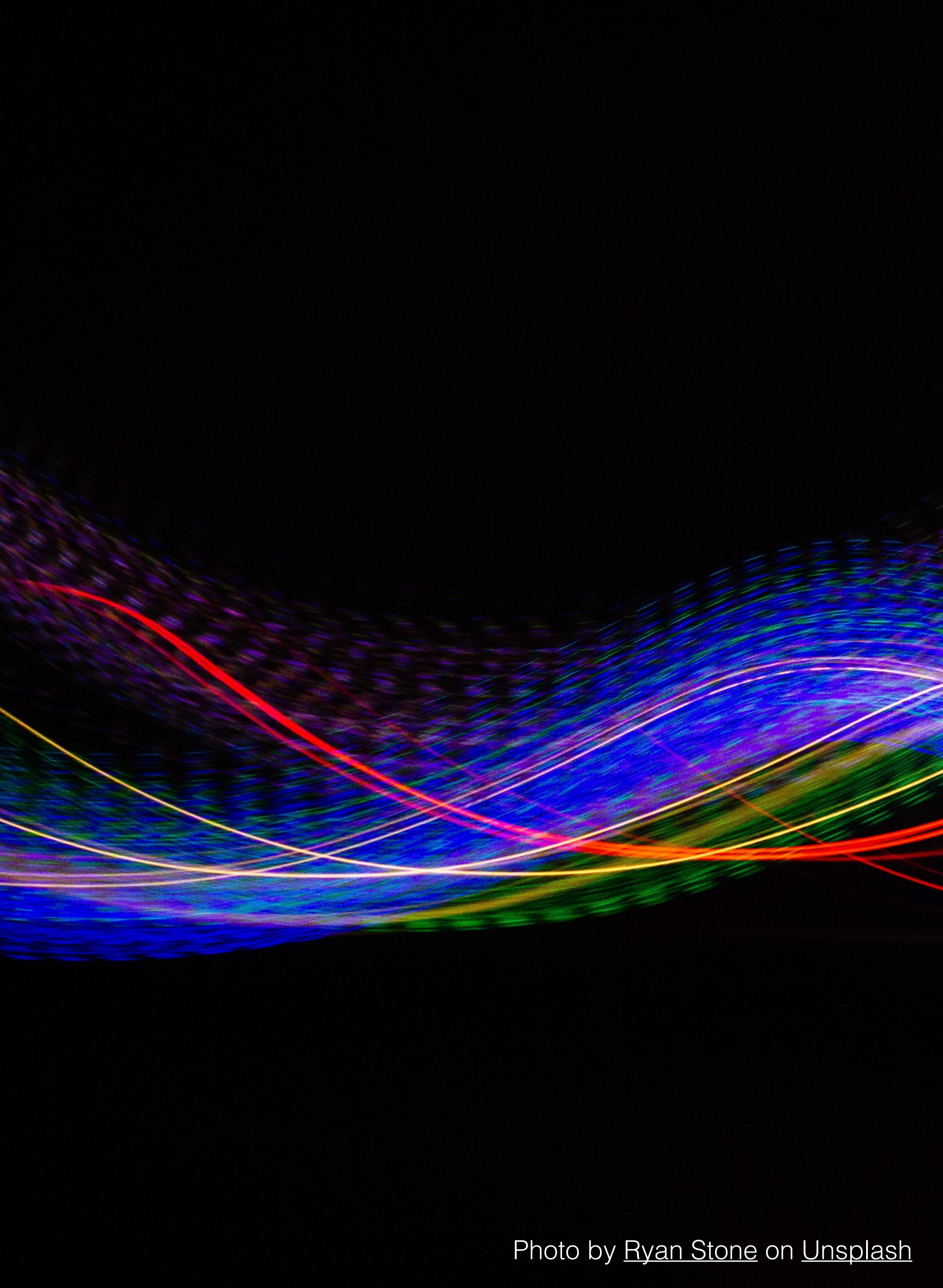
A **predictive model** is a statistical model used to provide a value (or range of values) of the response based on values of the predictors that were not used to train the model.



Like frequentist methodology, Bayesian methodology can be employed for explanation or prediction.

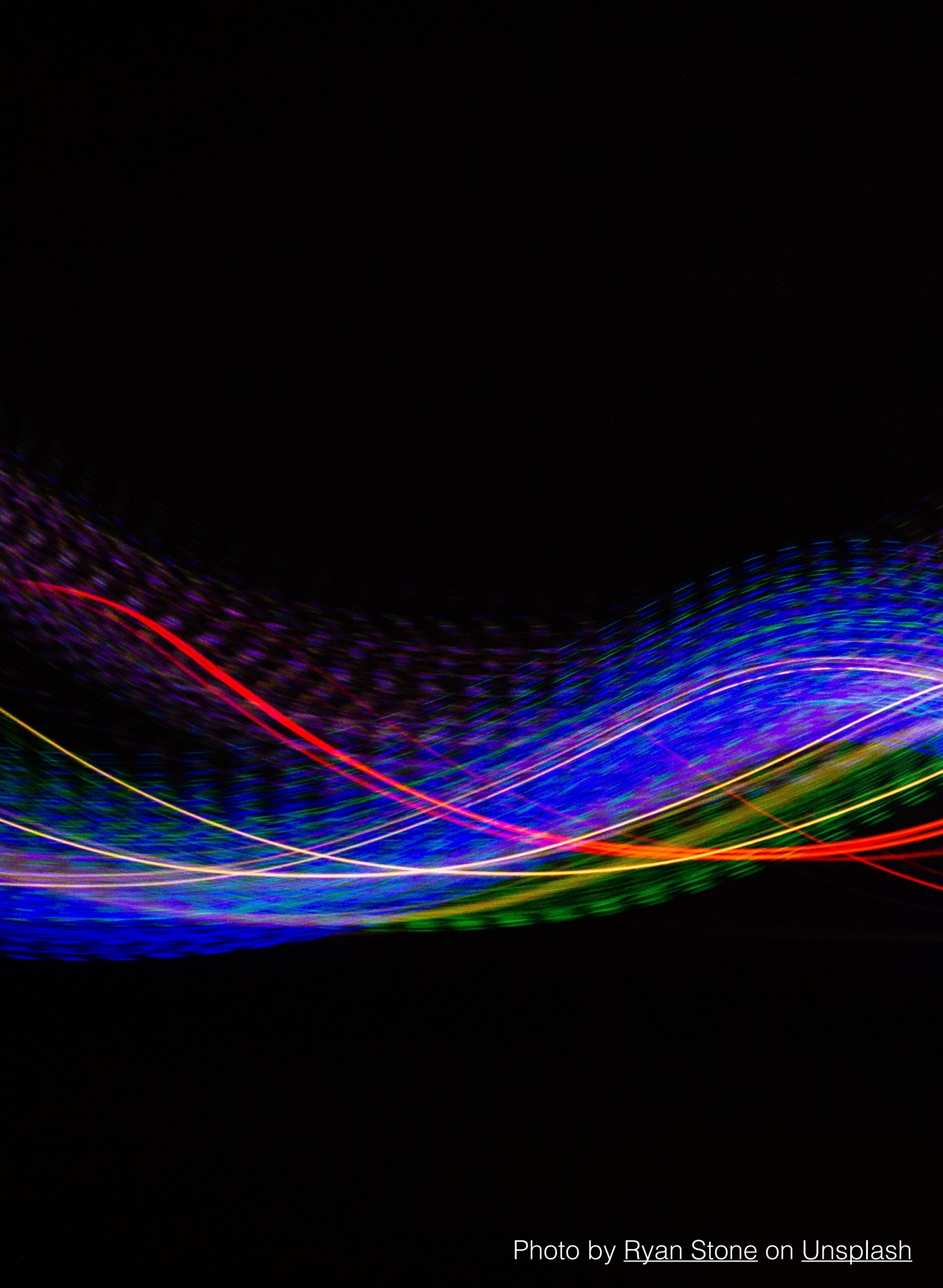
Let $(\mathbf{X}, f(\mathbf{x}; \theta))$ be a *frequentist* statistical model, and let $\mathbf{X}^*$ be a future observation that we would like to predict. Assume $\mathbf{X}^*$ is independent of $\mathbf{X}$ (e.g., we don't have a time series).

In frequentist inference, we would use an estimate of θ , denoted $\hat{\theta}$, and the standard error of $\hat{\theta}$, to derive point and interval estimates for $\mathbf{X}^*$.



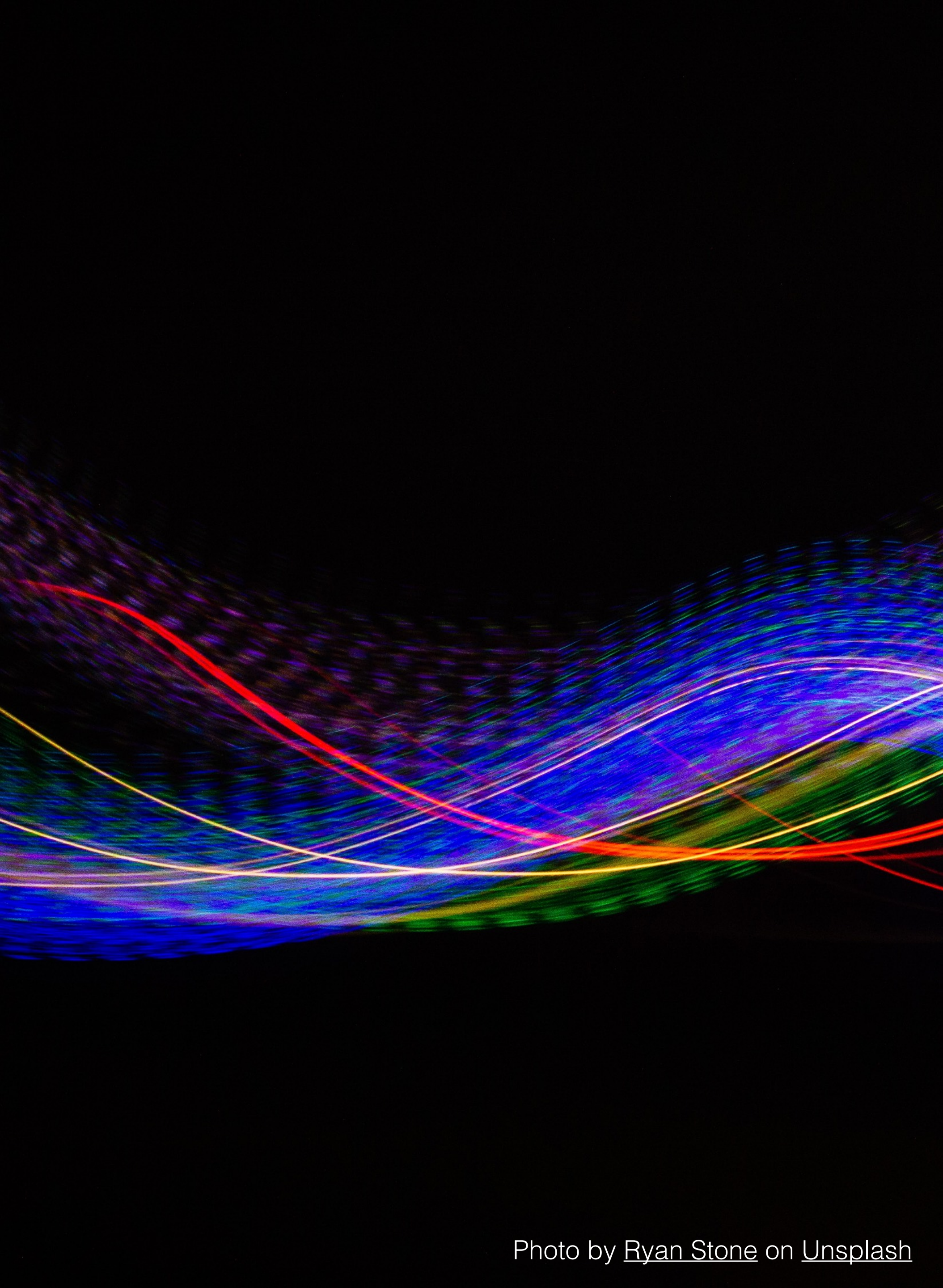
Let $(\mathbf{X}, f(\mathbf{x}, \theta))$ be our *Bayesian* statistical model, and let $\mathbf{X}^*$ be a future observation that we would like to predict. Assume $\mathbf{X}^*$ is independent of $\mathbf{X}$ (e.g., we don't have a time series).

Bayesian will derive a *distribution* of possible values for $\mathbf{X}^*$. This distribution will take into account the variability in the data generating process *and* variability in θ .



The **posterior predictive distribution**, $f(y^* \mid \mathbf{y})$, is the distribution of future observations, $\mathbf{X}^*$, given the observed data.

Derivation:



Example in R!